Supplemental Material

Bidirectional association between periodontal disease and diabetes mellitus: A systematic review and meta-analysis of cohort studies

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Table S1. Search term

a) Pubmed (all fields)
((periodontal OR periodontitis OR "periodontal disease" OR “clinical attachment loss”) AND (diabetes OR Hba1c OR HbA1c OR A1c OR glycated OR glycosylated OR glucose OR "Glycated Hemoglobin A"))
b) Web of Science (title, abstract, author keywords, keywords plus®)
TOPIC: (((periodontal OR periodontitis OR "periodontal disease" OR “clinical attachment loss”) AND (diabetes OR Hba1c OR HbA1c OR A1c OR glycated OR glycosylated OR glucose OR "Glycated Hemoglobin A")))

Table S2. Quality assessment using the Quality in Prognosis studies (QUIPS) tool.

Signalling question	**Authors' judgement for `yes**
Study participation: yes/no/unclear/NA (not applicable)	
a) Adequate participation in the study by eligible people	1) men and women ≥18 years without diabetes at baseline are eligible or 2) men and women ≥18 years without periodontal disease at baseline are eligible
b) Description of the source population or population of interest	Source population or population of interest is clearly described (e.g. region, age ...)
c) Description of the baseline study sample	Baseline study sample is clearly described
d) Adequate description of the sampling frame and recruitment	Recruitment, selection criteria and key characteristics of the source population are clearly described
e) Adequate description of the period and place of recruitment	Time period and place of recruitment are clearly described
f) Adequate description of inclusion and exclusion criteria	Inclusion and exclusion criteria are clearly described and presented.
Study participation: risk of bias rating (low/moderate/high/unclear)	**Low**: The study sample represents the population of interest and sufficient information on the recruitment and the selection process and the key characteristics of the population is available. **Moderate:** The study sample represents the population of interest but some information on the recruitment and/or the selection process and/or the key characteristics of the population were not described. **High:** The study sample does not represent the population of interest and/or insufficient information on the recruitment and the selection process and the key characteristics of the population is available. **Unclear:** not sufficient information were provided to judge this domain.
Study attrition: Yes/no/unclear/NA	
a) Adequate response rate for study participants	Response rate was adequate: ≥80% (<5 years) or ≥70% (≥5 years)
b) Attempts to collect information on participants who dropped out described	Attempts to collect information on participants who dropped out are described (e.g. telephone contact, mail, registers)
c) Reasons for loss to follow-up provided	Information about the reason participants were lost to follow-up are set up (e.g. deceased participants, participants refused or moved)
d) Adequate description of participants lost to follow-up	Key characteristics of participants lost to follow-up are described (Age, sex, diabetes status/ periodontal status at baseline)
e) No important differences between participants who completed the study and those who did not	No important differences in baseline characteristics between responders and non-responders

Study attrition: risk of bias rating (low/moderate/high/unclear) **Note:** In this domain, the first item was rated as the most important and thus decisively weighted	**Low:** If loss of follow-up was low and/or loss to follow-up was not associated with key characteristics and no potential bias of the observed association between the prognostic factor and the outcome is expected. **Moderate:** If loss of follow-up was low but not sufficient information were provided to judge if remaining loss of follow-up was associated with key characteristics, and thus, potential bias of the observed association between the prognostic factor and the outcome cannot be ruled out completely. **High:** If loss of follow-up was high and loss to follow-up was associated with key characteristics and potential bias of the observed association between the prognostic factor and the outcome is expected. **Unclear:** not sufficient information were provided to judge this domain.
Prognostic factor measurements: yes/no/unclear/NA	
a) Clear definition or description provided	1) Clear definition of periodontitis has been set by clinical oral examination (e.g. measurements of PPD or CAL) or 2) Clear definition of diabetes mellitus has been set by blood laboratory measurements (e.g. FPG, HbA1c values)
b) Adequately valid and reliable method of measurement	1) Valid and reliable methods of assessment for periodontitis include measurements of CAL or PPD and can be presented by Community Periodontal Index; We consider self-reported diagnosis that is validated by a physician and ICD-Codes from health insurance data as partial valid and reliable; Percentage of radiographic bone loss or self-reported periodontal disease or self-reported symptoms of periodontal disease without validation are not reliable; Technique for measurements of periodontal status is well described and measurements were performed by a dentist 2) Measurements of blood glucose or HbA1c levels are valid and reliable methods of measurement for diabetes mellitus; Self-reported history of diabetes with validation of the diagnosis by a physician and information from medical records are partial reliable; Only self-reported diagnosis without validation by a physician is not considered a valid and reliable method of measurement; Technique for laboratory measurements is well described.
c) Continuous variables reported or appropriate cut points used	1) Standard cut points for diagnosis of periodontitis: Probing pocket depth ≥ 4mm which corresponds to Community Periodontal Index code ≥ 3;

	clinical attachment loss: 1-2mm (mild); 3-4mm (moderate); ≥5mm (severe) or 2) Standard cut points for diagnosis of diabetes mellitus: Fasting plasma glucose ≥ 126 mg/dL or HbA1c >6,5%.
d) Same method and setting of measurement used in all study participants	1) Measurements of periodontal status are the same for all study participants 2) Measurements of glycemic status are the same for all study participants.
e) Adequate proportion of the study sample had complete data	Number in final model with complete data
f) Appropriate methods of imputation were used for missing data	Complete case analysis for exposure and outcome variables are required. If the analysis was almost complete case analysis, imputation was not required
Prognostic factor measurements: risk of bias rating (low/moderate/high/unclear)	**Low:** The prognostic factor of interest is adequately defined and measured in an adequate proportion of the study sample to sufficiently limit potential bias. **Moderate:** The prognostic factor of interest is partly adequately defined and measured in an adequate proportion of the study sample **High:** The prognostic factor of interest is not adequately defined and/or measured and/or missing in a substantial proportion of the study participants leading to potential bias. **Unclear:** not sufficient information were provided to judge this domain.
Outcome measurement: yes/no/unclear/NA	
a) Clear definition of the outcome provided	1) Clear definition for the diagnosis of diabetes mellitus has been set (e.g. FPG levels, HbA1c values) 2) Clear definition for the diagnosis of periodontitis has been set (e.g. PPD or CAL levels)
b) Use of adequately valid and reliable methods of outcome measurement	1) Valid and reliable methods of assessment for periodontitis include measurements of CAL or PPD and can be presented by Community Periodontal Index; We consider self-reported diagnosis that is validated by a physician and ICD-Codes from health insurance data as partial valid and reliable; percentage of radiographic bone loss or self-reported periodontal disease without validation are not reliable; Technique for measurements of periodontal status is well described 2) Measurements of blood glucose or HbA1c levels are valid and reliable methods of measurement for diabetes mellitus; Self-reported history of diabetes with validation of the diagnosis by a physician and information from medical records are partial reliable; Only self-reported diagnosis without validation by a physician is not considered a valid and reliable method of measurement;

	technique for laboratory measurements is well describe
c) Use of same method and setting of outcome measurement in all study participants	1) Measurements of type 2 diabetes mellitus are the same for all study participants or 2) Measurements of periodontal disease are the same for all study participants
Outcome measurement: risk of bias rating (low/moderate/high/unclear)	**Low:** The outcome of interest is adequately defined and measured in an adequate proportion of the study sample to sufficiently limit potential bias. **Moderate:** The outcome of interest is partly adequately defined and measured in an adequate proportion of the study sample **High:** The outcome of interest is not adequately defined and/or measured and/or missing in a substantial proportion of the study participants leading to potential bias. **Unclear:** not sufficient information were provided to judge this domain.
Study confounding: yes/no/unclear/NA	
a) Measurement of all important confounders	Minimal adjusted models should include: Age, sex, socio-economic status, smoking and BMI as most important confounders
b) Provision of clear definitions of the important confounders measured	Measurements of confounders are described and defined
c) Adequately valid and reliable measurements of all important confounders	Measurements of confounding factors are valid and reliable (e.g. validated questionnaires, examination by trained personal, etc.)
d) Use of the same method and setting of confounding measurement in all study participants	The method and setting of confounding measurement are the same for all participants
e) Appropriate imputation methods used for missing confounders (if applicable)	Appropriate methods of imputation for missing covariate data are applied and described.
f) Important potential confounders were accounted for the study design	Important potential confounders are accounted for in the study design (e.g., matching for key variables, stratification, use of multivariable analysis etc.).
g) Important confounders were accounted for in the analysis	Important potential confounders are accounted for in the analysis (i.e., appropriate adjustment).
Study confounding measurement: risk of bias rating (low/moderate/high/unclear) **Note:** Confounding is expected in all observational studies, because residual confounding cannot be completely excluded. Thus, for the item a) no study was assigned with “yes” and the total domain cannot be rated higher than moderate	**Low:** Important potential confounders were appropriately accounted for, and no potential bias of the observed association between the prognostic factor and the outcome is expected. **Moderate:** Studies adjusted for the defined minimal adjustment set and confounders were adequately defined and measured to sufficiently limit potential bias. **High:** Studies did not adjust for the defined minimal adjustment set and/or confounders were adequately defined and measured leading to potential bias. **Unclear:** not sufficient information were provided to judge this domain.

Statistical analysis and reporting: yes/no/unclear/NA	
a) Sufficient presentation of data to assess the adequacy of the analytic strategy	There is sufficient presentation of data to assess the adequacy of the analysis (e.g. findings are displayed in a table or in the text). The results are expressed as risk ratios (e.g. hazard ratios, relative risks, odds ratios) with corresponding 95% confidence intervals
b) Strategy for model building is appropriate and based on conceptual framework model	The selection of the confounders are described and appropriate (e.g. literature-based); step-wise regression is not appropriate
c) Statistical model is adequate for the study design	Multivariable logistic regression or cox proportional hazard model are applied. Univariate methods are not appropriate.
d) No selective reporting of results	There is no selective reporting of results (e.g. findings are shown for a specify age group, time period etc.)
Statistical analysis and reporting: risk of bias rating (low/moderate/high/unclear)	**Low:** The statistical analysis was appropriate and the data are sufficiently reported. **Moderate:**. The statistical analysis was appropriate but the strategy for model building is not adequately described and/or for the main findings there is no selective reporting of the data. **High:** The statistical analysis was not appropriate and/ or there is selective reporting of the data. **Unclear:** not sufficient information were provided to judge this domain.

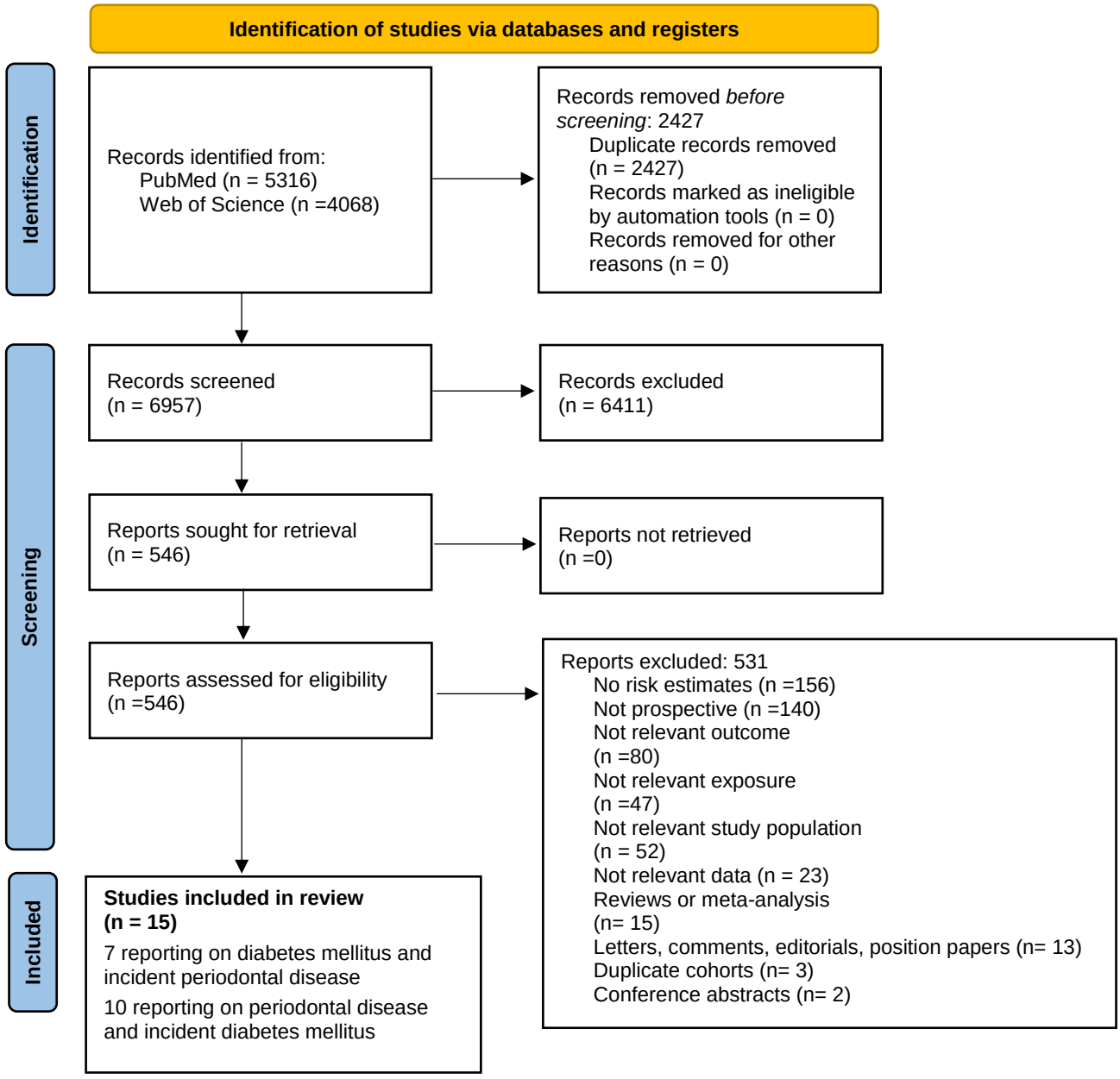

Figure S1. Flow chart of the study selection process

Template from: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71.

Table S3. Excluded studies based on full-text screening

Exclusion criteria	**Excluded studies**
No risk estimates (studies that do not have a risk ratio of the association of periodontal disease and diabetes mellitus)	[1-156]
Not prospective (cross-sectional studies, case-controls studies, no prospective associations)	[157-296]
Not relevant outcome (studies that do not have an indicator of periodontal disease or diagnosis of diabetes mellitus as outcome)	[297-376]
Not relevant study population (e.g. studies in children and adolescents, pregnant women)	[377-428]
Not relevant exposure (studies that do not have an indicator of periodontal disease or diagnosis of diabetes mellitus as exposure)	[429-475]
Not relevant data (studies that do not have a relevant exposure and outcome)	[476-498]
Systematic reviews/Meta-analysis	[499-513]
Letter, comments, editorials, position paper	[514-526]
Duplicate cohorts	[527-529]
Conference abstracts	[530, 531]

Table S4. Study characteristics of cohort studies included in the meta-analysis

First author, publication year, country	Study design, Study name or description, Follow-up	Sex, age	Number of participants, number of cases	Exposure, exposure assessment	Exposure categories	Outcome, outcome assessment	Relative Risk (95% CI)	Adjustment for confounders
Studies of diabetes mellitus and incident periodontal diseases								
Alshihayb et al., 2020 USA[532]	Prospective cohort study, Veterans Affairs (VA) Dental Longitudinal Study (DLS) is a subset of the VA Normative Aging Study (NAS), 14 y	Men, 58.9y	521, 264	Type 2 diabetes, existing NAS medical records	Diabetes yes/no	Incident periodontal Disease (severe), Clinical periodontal examinations: probing pocket depth at multiple sites per tooth & attachment loss	1.32 (0.77; 2.26)	Time-varying age, education, time-varying body mass index, time-varying comprehensive smoking index
Chiu et al., 2015, Taiwan[533]	Prospective cohort study, Keelung community-based integrated screening (KCIS) program, 5 y	Women and men, 35-44 y	4387, 1247	Type 2 diabetes, Fasting plasma glucose (FPG) or self-reported type 2 diabetes	FPG normal (<100 mg/dL), Type 2 diabetes (≥126 mg/dL)	Incident periodontal disease, measurement of periodontal pocket depth (PPD) and classified by Community Periodontal Index (CPI)	1.95 (1.22; 3.13)	Age, gender, years of education, betel nut chewing, smoking, alcohol consumption, waist, triglyceride, HDL, elevated blood pressure, fruit intake, physical activity
Jimenez et al., 2012, USA[534]	Prospective cohort study Health Professionals Follow-Up Study (HPFS), 20 y	Men, 40-75 y	35247, 3009	Type 2 diabetes, self-reported: 1) ≥1classic symptom of diabetes and FPG of ≥140mg/dl (cases reported after 1996: ≥ 126mg/dl), non-FPG	Diabetes yes/no	Periodontal disease, Self-reported	1.29 (1.13; 1.47)	Age, gender, smoking status, BMI, fruit/vegetable intake, number of teeth at baseline, physical activity, alcohol

				≥200mg/dl; or 2) elevated plasma glucose concentrations on at least two different occasions in the absence of symptoms; or 3) hypoglycemic				consumption, dental profession, baseline history of stroke, coronary artery bypass surgery or myocardial infarction
Morita et al., 2012, Japan[535]	Prospective cohort study, Participants in this study worked in and around Nagoya City and had annual health check-ups between December 1997 and February 2006, 5 y	Women and men, 30-69 y	5856, 2068	HbA1c level, measured in a fasting venous blood sample	Diabetes: no (HbA1c < 6,5%), Diabetes: yes (HbA1c ≥ 6,5%)	Periodontal disease, The PPD around 10 index teeth were examined to obtain the CPI	1.17 (1.01; 1.36)	Age, gender, smoking status, BMI
Lee et al., 2018, South Korea[536]	Prospective cohort study, National Health Insurance Service-Elderly Cohort (NHIS-EC), 11 y	Women and men, ≥ 60 y	149785, N.A.	Type 1 and 2 diabetes, ICD-10 codes: E101, E105, E109, E111, E115, E119, E131, E135, E139, E141, E145, E149	Diabetes yes/no	Periodontal disease, based on the guidelines of the Centers for Disease Control and American Academy of Periodontology using dental history questionnaires, clinical signs, and an oral examination and radiographic evaluation: diagnosis of PD by ICD-10 ([K052],[K053],[K054], [K055], [K056]) and who received periodontal treatment	1.18 (1.17; 1.20)	Age, gender, household income, insurance status, residence area, health status and other chronic diseases
Lee, et al., 2019,	Prospective cohort study, National Health	Women and men,	78768, 15170	Mainly type diabetes 2, ICD-9	Mainly type 2 diabetes yes/no	Periodontal Disease, ICD-9 code:	1.04 (1.01; 1.08)	Age, sex, income, dyslipidemia,

Taiwan[498]	Insurance database Taiwan (NHIRD), Approx. 9 y	>20 y		codes 250 and 251, but diagnosis after age of 20 years		periodontitis (523.0 and 523.5)		hypertension, rheumatoid arthritis, Charlson comorbidity index
Sun et al., 2018, Taiwan[537]	Prospective cohort study, National Health Insurance database Taiwan (NHIRD), 14 y	Women and men, <40 y	21240, 794	Type 1 diabetes, ICD-9 codes 250.x1 and 250.x3)	Type 1 diabetes yes/no	Periodontal disease, ICD-9 code: periodontitis (523.3 and 523.4)	1.66 (1.41; 1.96)	Age, gender, urbanization level, comorbidities of CAD, stroke, asthma, COPD, alcohol-related disease and mental disease
Studies of periodontal disease and incident diabetes								
Alshihayb et al., 2020, USA[532]	Prospective cohort study, Veterans Affairs (VA) Dental Longitudinal Study (DLS) is a subset of the VA Normative Aging Study (NAS), 14 y	Men, 58.9y	672, 82	Periodontal disease Clinical periodontal examinations: probing pocket depth at multiple sites per tooth & attachment loss	No/mild vs. moderate/severe periodontal disease	Type 2 diabetes, medical examination interviews based on self-report	1.33 (0.71; 2.52)	Time-varying age, education, time-varying body mass index, time-varying comprehensive smoking index .
Lin et al., 2014, Taiwan[538]	Retrospective cohort study, National Health Insurance Research Database (NHIRD), 5.47 y	Women and men, ≥ 53 y	44601, 2501	Periodontal disease, based on claims data (ICD-9-CM codes 523.4 and 523.5); needing subgingival curettage (procedure codes 91006C, 91007C, and 91008C) and periodontal flap procedure (procedure codes 91009C and 91010C)	Periodontitis needing surgical treatment no = comparison / yes = severe Periodontitis	Type 2 diabetes, patients who have been diagnosed with ICD-9-CM codes 250 at least two times and concomitantly received antidiabetic medications	1.19 (1.10; 1.29)	Age, gender, income, urbanization, hypertension, CAD, hyperlipidemia, obesity

Demmer et al, 2008, USA[539]	Prospective cohort study, National Health and Nutrition Examination Survey (NHANES I) and its Epidemi- ologic Follow-up Study (NHEFS), 17 y	Women and men, 25-74 y	9296, 817	Periodontal disease, measurement of gingival inflammation extent, the presence or absence of periodontal pockets, tooth mobility and classified by Periodontal Index (PI)	Category PI0: periodontal healthy (PI= 0); Category PI1: gingivitis (0< PI≤ 0.87); Category PI2: periodontitis (0.88≤ PI≤ 1.60); Category PI3: periodontitis (1.61≤ PI≤ 2.44); Category PI4: periodontitis (2.45≤ PI≤ 5.07); Category PI5: periodontitis (5.08 ≤ PI≤ 8.0); Edentulous	Type 2 diabetes, diabetes listed on the death certificate, self-reported physician diagnosis requiring pharmacological treatment, health care facility stay with a discharge diagnosis of diabetes	PI2: 1.03 (0.65; 1.64) PI3: 2.08 (1.51; 2.87) PI4: 1.71 (1.19; 2.45) PI5: 1.50 (0.99; 2.27) pooled to 1.64 (1.35, 1.99)	Age, gender, race, education, smoking status, BMI, physical activity, hypertension, total cholesterol, subscapular skinfold
Ide et al, 2011, Japan[540]	Prospective cohort study, worksite cohort study, 6.5 y	Women and men, 30-59 y	5848, 287	Periodontal disease, Community Periodontal Index (CPI)	no pathological pockets (CPI score: 0,1 or 2), moderate periodontitis (CPI score: 3), severe periodontitis (CPI score: 4)	Type 2 diabetes, FPG > 125 mg/dL	CPI score 3: 1.00 (0.77; 1.30) CPI score 4: 1.28 (0.89; 1.86) pooled to 1.09 (0.88, 1.35)	Age, gender, smoking status, BMI, triglyceride level, hypertension, high density lipoprotein level, gamma-glutamyl transpeptidase

Miyawaki et al.,2016, Japan[541]	Prospective cohort study, MY Health Up study, 5 y	Men, 36-55 y	2469, 133	Periodontal status, Self-reported periodontal symptoms	Tooth loosening no/yes	Type 2 diabetes, self-reported or met blood test criteria at least once during the follow-up period (FPG level ≥126 mg/dl between 2005 and 2009 and/or HbA1c values ≥ 6,5% in 2008 and 2009)	1.73 (1.14; .,64)	Age, current smoking habits, BMI, family history of diabetes, hypertension, alcohol heavy consumption (≥ 40 g/day), exercise habits (> 30min, ≥ 2 days/week)
Kebede et al.,2012, Germany[542]	Study of Health in Pomerania: SHIP-0 (baseline) and SHIP-2 (follow-up), 11.1 y	Women and men, 56.8 y	2034, 206	Periodontal disease, Periodontal measurements assessed at four sites per tooth according to the half-mouth method: 1) PPD 2) CAL	1) Mean PPD: analyzed as quartiles and continuously, 2) Mean clinical attachment level analyzed in quartiles and continuously	Diabetes, known diabetes (self-reported physician diagnoses or treatment with antidiabetic medication) or HbA1c levels ≥ 6.5% or non-FPG levels ≥11,1 mmol/L	1) Q2: 1.35 (0.82; 2.32) Q3: 1.22 (0.75; 2.00) Q4: 1.27 (0.78; 2.07) 2) Q2: 0.61 (0.36; 1.04) Q3: 0.92 (0.56; 1.49) Q4: 0.82 (0.49; 1.37) pooled to 0.78 (0.88, 1.05)	Age, gender, highest level of general education, marital status, waist circumference, physical activity, smoking status, dental visits past 12 months, follow-up time

Myllymaki et al.,2018, Finland[543]	Prospective cohort study, MY Health up study, 5 y	Women and men, 36-55 y	394, 81	Periodontal condition, Clinical oral examinations by 2 dentists: presence of deepened periodontal pockets (4mm or deeper)	periodontally healthy = No deepened periodontal pockets, PPD of 4-5 mm, PPD of ≥ 6 mm, edentulous, Number of sites with periodontal pockets 4 mm deep or deeper (continuous variable)	Type 2 diabetes, fasting venous plasma glucose ≥7,0 mmol/L and/or 2-hour venous plasma glucose ≥ 11,1 mmol/L after ingestion of 75g of an oral glucose load	PPD of 4-5 mm: 1.32 (0.69; 2.53) PPD of ≥ 6 mm: 1.56 (0.58; 1.92) pooled to 1.44 (0.93, 2.26)	Gender, risk of diabetes mellitus, physical activity, dietary habits, IGT at baseline, hypertriglyceridemia, low HDL-C, smoking status, BMI, absolute change in BMI during follow up time
Winning et al.,2017, Northern Ireland[544]	Prospective cohort study, PRIME study (Prospective Epidemiological Study of Myocardial Infarction), 7.8 y	Men, 58-72 y	1331, 80	Periodontal disease, clinical periodontal measurements: 1) PPD 2) CAL	no/mild periodontitis, moderate Periodontitis: ≥2 interproximal sites with CAL ≥4 mm, not on the same tooth or ≥2 interproximal sites with PPD ≥5 mm, not on the same tooth, severe periodontitis: ≥ 2 interproximal sites with CAL ≥6 mm, not on the same tooth, and ≥1 interproximal sites with PPD ≥5 mm	Type 2 diabetes, 1)self-reported: diabetes diagnosis, listing a medication that suggested diabetes management and 2) validated by general medical practioner: FPG ≥ 126mg/dl	Moderate periodontitis: 1.53 (0.86; 2.74) Severe periodontitis: 1.85(1.06; 3.22) pooled to 1.69 (1.13, 3.22)	Age, number of teeth, smoking status, toothbrushing frequency, marital status, baseline BMI, baseline CRP, cholesterol, history of CAD, history of hypertension, education years, dental attendance, socio-economic status

Morita et al., 2012, Japan[535]	Prospective cohort study, Participants in this study worked in and around Nagoya City and had annual health check-ups between December 1997 and February 2008, 5 y	Women and men, 30-69 y	6125, 168	Periodontal disease, The PPD around 10 index teeth were examined to obtain the CPI	CPI Code 0, CPI Code 3, CPI Code 4	Type 2 diabetes (HbA1C ≥6,5%), measured in a fasting venous blood sample	CPI score 3: 2.47 (0.78; 7.79) CPI score 4: 3.45 (1.08; 11.02) pooled to 2.91 (1.29, 6.60)	Age, gender, BMI, alcohol consumption, smoking status
Lee et al., 2017, South Korea[545]	Prospective cohort study, National Health Insurance Service-Health Examinee Cohort (NHIS-HEC), 12 y	Women and men, 40-79 y	354850, 110006	Periodontal disease, diagnosed clinically in oral checkup examinations in accordance with criteria of the Centers for Disease Control and Prevention/American Academy of Periodontology by a general dentist or a periodontitis (KCD-6 codes K05.2–K05.6, corresponding to ICD-10 codes K05.2–K05.6)	Oral healthy, Periodontal disease	Type 1 and 2 diabetes, diagnosed by physicians or other medical professionals: diabetes mellitus (KCD-6 codes E10–E14, corresponding to ICD-10 codes E10–E14)	1.16 (1.12; 1.02)	Age, gender, smoking status, income, insurance status, residence area

Table S5. Risk of bias assessment of all included studies in the meta-analysis using the Quality in Prognosis studies (QUIPS) tool.

	Domains of bias: Low, moderate, high risk of bias, unclear						
Author, year	Study participation	Study attrition	Prognostic factor measurement	Outcome measurement	Study confounding	Statistical analysis and reporting	**Overall**
Studies of diabetes mellitus and incident periodontal disease							
Alshihayb, 2020[532]	Moderate	Low	High	Low	Moderate	Low	High
Chiu, 2015[533]	Low	Moderate	Low	Low	Moderate	Low	Low
Jimenez, 2012[534]	Low	Low	Moderate	High	Moderate	Low	High
Lee, 2018[536]	Moderate	Unclear	Moderate	Low	High	Low	Moderate
Lee, 2019[546]	Low	Unclear	Moderate	Moderate	High	Low	Moderate
Morita 2012[535]	Low	High	Low	Low	Moderate	Low	Moderate
Sun, 2018[537]	Low	Unclear	Moderate	Moderate	High	Low	Moderate
Studies of periodontal disease and incident diabetes mellitus							
Alshihayb, 2020[532]	Moderate	Low	Low	High	Moderate	Low	High
Demmer, 2008[539]	Low	Low	Moderate	Moderate	Moderate	Low	Moderate
Ide, 2011[540]	Low	Moderate	Low	Low	Moderate	Low	Low
Kebede, 2018[542]	Low	High	Low	Low	Moderate	Low	Low
Lee, 2017[545]	Low	Low	Low	Low	Moderate	Low	Low
Lin, 2014[538]	Low	Unclear	High	Moderate	High	Low	High
Miyawaki, 2016[541]	Low	Moderate	High	Low	Moderate	Low	High
Morita, 2012[535]	Low	High	Low	Low	Moderate	Low	Moderate
Myllymaki, 2018[543]	Moderate	High	Low	Low	Moderate	Low	Moderate
Winning, 2017[544]	Low	Low	Low	Low	Moderate	Low	Low

Table S6: Certainty of evidence by applying the GRADE tool

Certainty assessment							Relative risk (95% CI)	Certainty
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations		
Periodontal diseases and incidence of diabetes mellitus								
10	observational studies	not serious	serious[a]	serious[b]	not serious	Dose response gradient	**RR 1.26** (1.12 to 1.41)	⨁⨁⨁◯ MODERATE
Diabetes mellitus and incidence of periodontal diseases								
7	observational studies	not serious	not serious[c]	serious[b]	not serious	none	**RR 1.24** (1.13 to 1.37)	⨁⨁⨁◯ MODERATE

[a] RR ranges from 0.78 to 2.91 , and 95% CIs do not overlap; I^2: 71%

[b] different assessment methods were used to diagnose the disease (e.g. defined by self-reports, clinical examinations or a combination of both), some cohorts include participants with type 1 and type 2 diabetes, different time of follow-up

[c] I^2 high (92%), but RRs in same direction

Table S7: Summary relative risks (SRRs) and 95% confidence intervals (95% CIs) of periodontal diseases and incidence of diabetes mellitus by subgroups

	n of studies	SRRs (95% CIs)	I^2 (%)	P_{within}	$P_{between}$
All studies	10	1.26 (1.12, 1.41)	71	<0.0001	
Sex[a]					
Men	5	1.31 (1.01, 1.70)	58	0.048	0.109
Women	2	2.00 (1.58, 2.54)	0	0.974	
Risk of bias					
Low	4	1.09 (0.90, 1.32)	67	0.028	0.127
Moderate	3	1.66 (1.35, 2.03)	10	0.331	
High	3	1.30 (1.04, 1.63)	34	0.219	
Type of diabetes					
Type 1 diabetes	-	-	-	-	0.080
Type 2 diabetes	8	1.41 (1.19, 1.67)	63	0.009	
Type 1 + type 2 diabetes	2	0.98 (0.67, 1.44)	85	0.009	
Assessment of diabetes					
Low risk of bias	7	1.24 (1.03, 1.50)	67	0.005	0.951
Moderate risk of bias	2	1.38 (1.01, 1.89)	89	0.002	
High risk of bias	1	1.33 (0.71, 2.51)	no meta-analysis n=1		
Assessment of periodontal diseases					
Low risk of bias	7	1.19 (0.99, 1.42)	60	0.019	0.613
Moderate risk of bias	1	1.64 (1.35, 1.99)	no meta-analysis n=1		
High risk of bias	2	1.35 (0.96, 1.92)	66	0.086	
Geographical location					
Asia	5	1.19 (1.09, 1.30)	55	0.063	0.636
USA	2	1.61 (1.34, 1.94)	0	0.535	
Europe	3	1.20 (0.72, 1.99)	78	0.012	
Duration of follow-up					
<10 years	5	1.34 (1.09, 1.65)	60	0.043	0.471
≥10 years	5	1.22 (0.96, 1.55)	80	<0.0001	

Number of cases					
Cases <1000	8	1.38 (1.08, 1.77)	73	<0.0001	0.495
Cases ≥1000	2	1.16 (1.13, 1.20)	0	0.564	
By smoking status[b]					
Never	3	1.39 (1.10, 1.76)	7	0.340	0.718
Ever	2	1.28 (1.10, 1.66)	78	0.033	
Adjustment for education					
Yes	4	1.29 (0.83, 1.99)	83	<0.0001	0.865
No	6	1.20 (1.10, 1.31)	49	0.082	
Adjustment for smoking status					
Yes	9	1.31 (1.10, 1.56)	74	<0.0001	0.703
No	1	1.19 (1.10, 1.29)	no meta-analysis n=1		
Adjustment for overweight					
Yes	9	1.32 (1.10, 1.58)	72	<0.0001	0.624
No	1	1.16 (1.12, 1.20)	no meta-analysis n=1		
Adjustment for fruit and vegetable intake					
Yes	-	-	-	-	-
No	10	1.26 (1.12, 1.41)	71	<0.0001	
Adjustment for alcohol intake					
Yes	2	1.98 (1.27, 3.11)	19	0.267	0.090
No	8	1.21 (1.09, 1.35)	70	0.002	
Adjustment for physical activity					
Yes	4	1.33 (0.89, 1.97)	84	<0.0001	0.921
No	6	1.18 (1.10, 1.27)	35	0.117	
Adjustment for number of missing teeth					
Yes	1	1.69 (1.00, 2.85)	no meta-analysis n=1		0.488
No	9	1.24 (1.10, 1.39)	76	<0.0001	

P_{within}, P for heterogeneity within each subgroup; $P_{between}$, P for heterogeneity between subgroups with meta-regression

[a] information available from 5 studies [532, 539-541, 544]

[b] information available from 3 studies [539, 540, 543]

Table S8: Summary relative risks (RRs) and 95% confidence intervals (95% CIs) of diabetes mellitus and incidence of periodontal diseases by subgroups

	n of studies	SRRs (95% CIs)	I^2 (%)	P_{within}	$P_{between}$
All studies	7	1.24 (1.13, 1.37)	92	<0.0001	
Sex[a]					
Men	3	1.35 (1.21, 1.51)	0	0.411	0.447
Women	1	1.57 (1.25, 1.98)	no meta-analysis n=1		
Risk of bias					
Low	1	1.95 (1.22, 3.12)	no meta-analysis n=1		0.404
Moderate	4	1.21 (1.08, 1.34)	95	<0.0001	
High	2	1.29 (1,14; 1.47)	0	0.935	
Type of diabetes					
Type 1 diabetes	1	1.66 (1.41, 1.96)	no meta-analysis n=1		0.306
Type 2 diabetes	4	1.25 (1.01, 1.55)	82	0.001	
Type 1 + type 2 diabetes	2	1.18 (1.17, 1.19)	no meta-analysis n=1		
Assessment of diabetes					
Low risk of bias	2	1.44 (0.88, 2.34)	76	0.043	0.917
Moderate risk of bias	4	1.23 (1.11, 1.37)	96	<0.0001	
High risk of bias	1	1.32 (0.77, 2.26)	no meta-analysis n=1		
Assessment of periodontal diseases					
Low risk of bias	3	1.36 (1.00, 1.86)	56	0.124	0.911
Moderate risk of bias	3	1.22 (1.08, 1.38)	97	<0.0001	
High risk of bias	1	1.29 (1.13, 1.47)	no meta-analysis n=1		
Geographical location					
Asia	5	1.23 (1.11, 1.37)	94	<0.0001	0.957
USA	2	1.29 (1.13, 1.47)	0	0.94	
Duration of follow-up					
<10 years	3	1.18 (0.97, 1.43)	78	0.011	0.484
≥10 years	4	1.34 (1.13, 1.58)	84	<0.0001	
Number of cases[b]					0.201

Cases <1000	2	1.63 (1.39, 1.91)	0	0.425	
Cases ≥1000	4	1.21 (1.03, 1.42)	83	<0.0001	
By smoking status[c]					
Never	1	1.32 (1.06, 1.64)	no meta-analysis n=1		-
Ever	1	1.30 (1.14, 1.49)	no meta-analysis n=1		
Adjustment for education					
Yes	2	1.64 (1.12, 2.40)	13	0.285	0.272
No	5	1.22 (1.11, 1.34)	94	<0.0001	
Adjustment for smoking status					
Yes	4	1.28 (1.12, 1.47)	32	0.211	0.706
No	3	1.22 (1.08, 1.38)	97	<0.0001	
Adjustment for overweight					
Yes	3	1.24 (1.12, 1.36)	0	0.611	0.746
No	4	1.25 (1.11, 1.42)	96	<0.0001	
Adjustment for fruit and vegetable intake					
Yes	2	1.49 (1.01, 2.18)	64	0.098	0.417
No	5	1.21 (1.09, 1.34)	94	<0.0001	
Adjustment for alcohol intake					
Yes	2	1.49 (1.01, 2.18)	64	0.098	0.417
No	5	1.21 (1.09, 1.34)	94	<0.0001	
Adjustment for physical activity					
Yes	2	1.49 (1.01, 2.18)	64	0.098	0.417
No	5	1.21 (1.09, 1.34)	94	<0.0001	
Adjustment for number of missing teeth at baseline					
Yes	1	1.29 (1.13, 1.47)	no meta-analysis n=1		0.994
No	6	1.24 (1.11, 1.37)	93	<0.0001	

P_{within}, P for heterogeneity within each subgroup; $P_{between}$, P for heterogeneity between subgroups with meta-regression

[a] information available from 3 studies[532], [534], [537]

[b] information available from 6 studies: one study did not provide number of cases in the text [536]

[c] information available from 1 study [534]

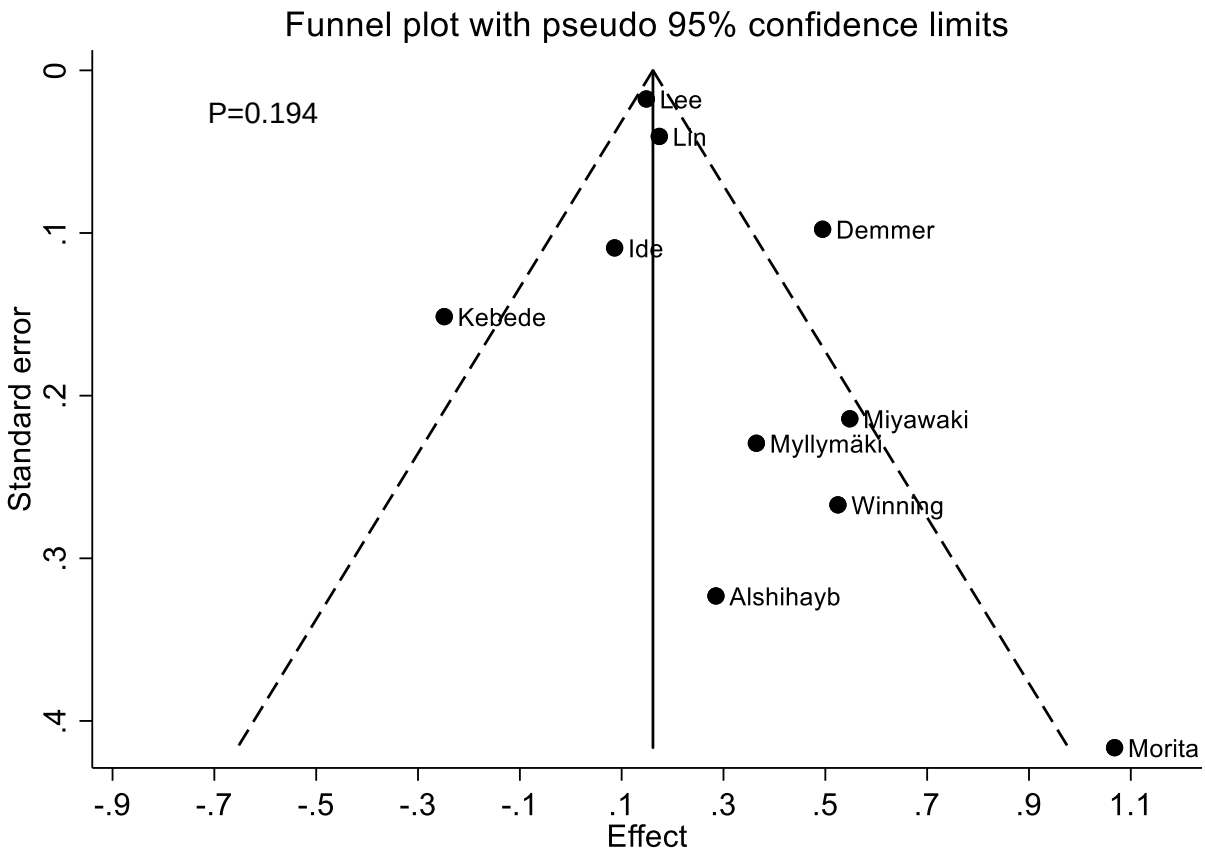

Figure S2. Funnel plot for association between periodontal disease and incidence of diabetes mellitus

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